

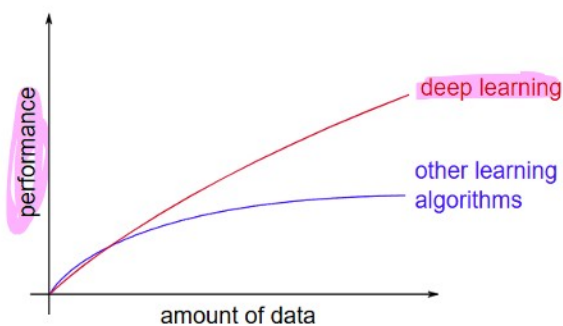
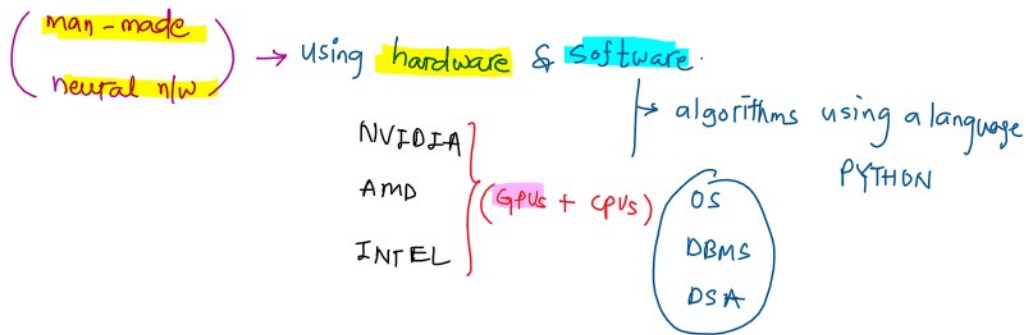
What is Deep Learning ?

Deep learning is a subset of machine learning (ML & DL are subsets of AI) that uses artificial neural network (ANN) to mimic the learning process of the human brain.

↓

ANN

Artificial Neural Network



[CPU vs GPU vs TPU
- wait for module #3]

- ANN is a computational model inspired by the structure and function of the brain.

↓

[there are billions of neurons]

- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

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AI terminologies are not only confusing but are buzzwords?

Disclaimer: No intention to undermine brand's identity.



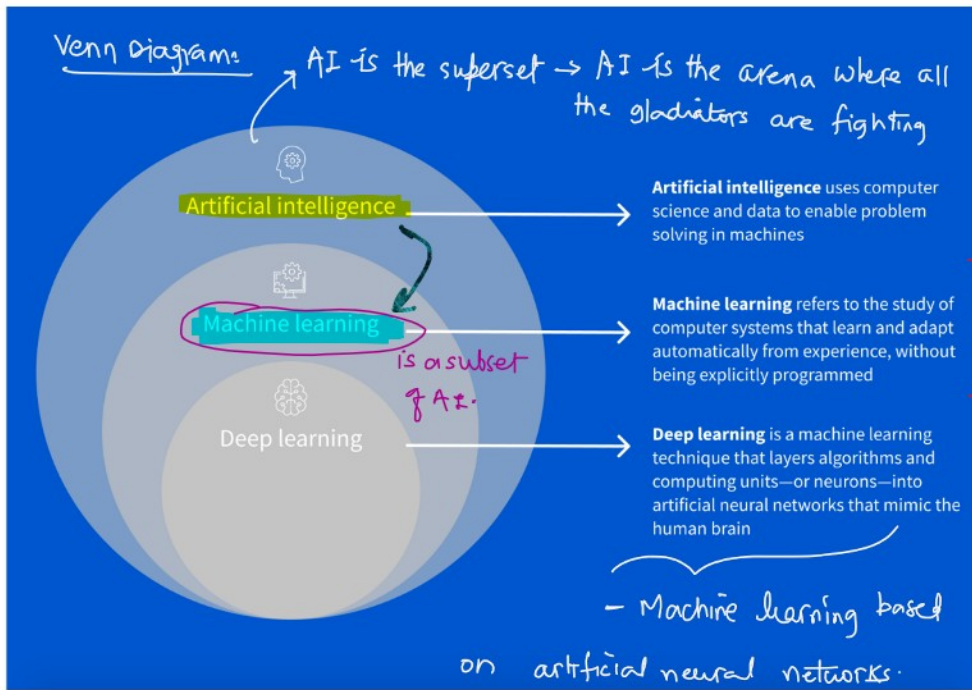
→ it doesn't mean AI here.

just a marketing gimmick.



AI is needed for sales & marketing.

Difference between AI, ML and DL



AI is the playground to engineer machines empowered by data to mimic cognitive functions

Data is the new oil (Human Intelligence)

Pro-Tip

Software development vs Machine learning

Aspect	Traditional Programming (Chef with strict recipe)	Machine Learning (ML Chef)
Approach	Follows a strict recipe (explicit rules)	Experiments and self-improves (self-adjusting rules)
Example Steps	1. Add 2 cups flour 2. Add 1 cup sugar 3. Bake at 180°C for 30 min	Bake a cake → Get feedback → Adjust next time
Consistency	Always produces the same result	Result improves over time based on feedback
Ability to Learn from Mistakes	Does not learn from past mistakes	Learns and adjusts based on feedback
Flexibility	Rigid, rule-based	Flexible, adaptive
Outcome over time	Same cake every time	Better cake over time
Nature of Rules	Explicit, hand-coded (IF-ELSE, loops)	Self-adjusting, data-driven
Core Analogy	Chef blindly following a recipe	Chef who learns and adapts based on experience

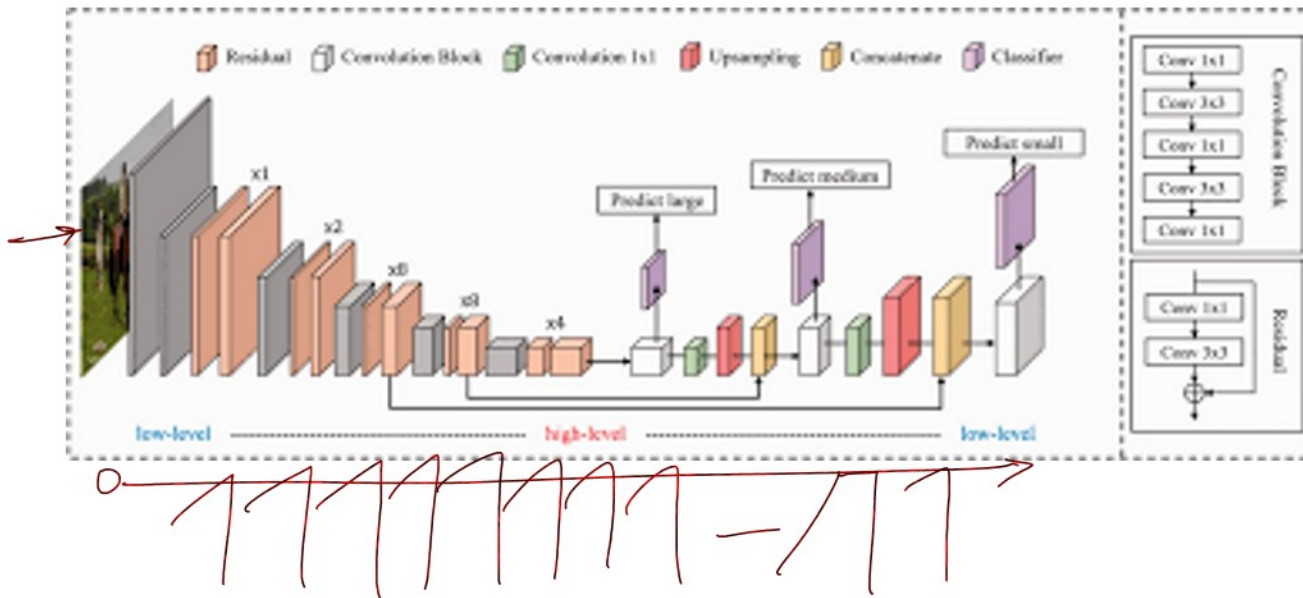
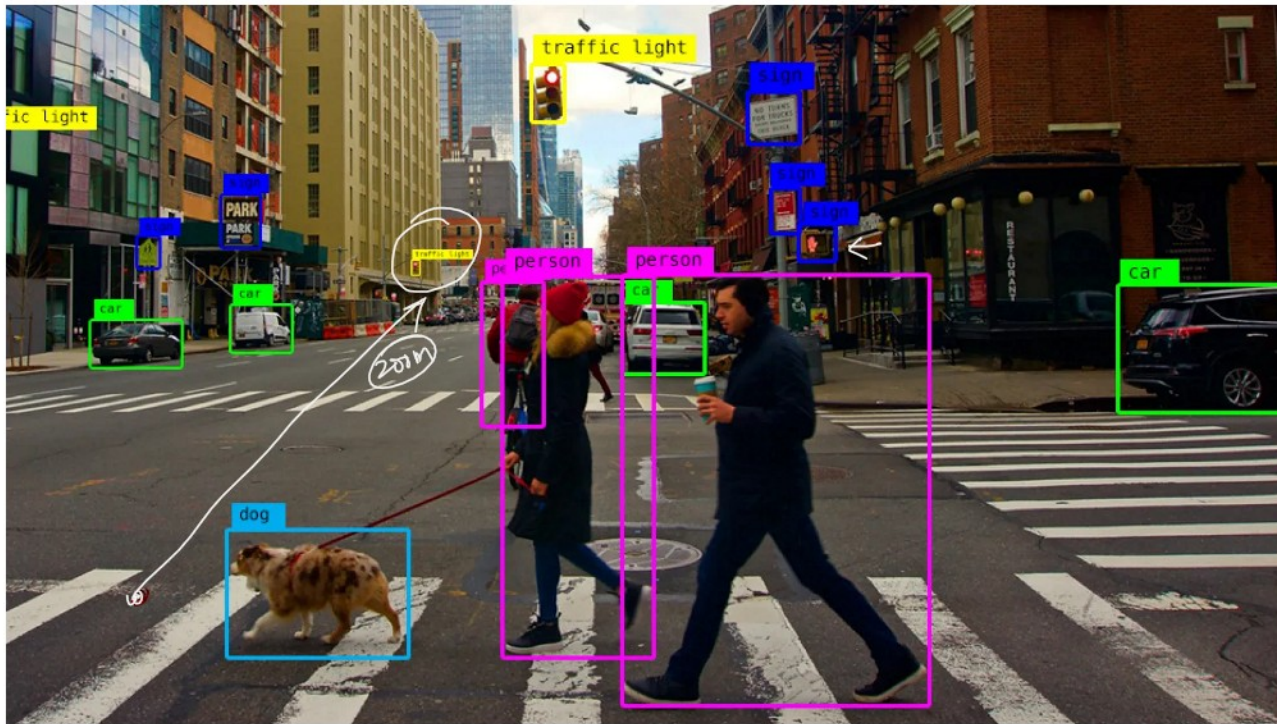
Pro-Tip

What is 'deep' in the deep learning?

DL is a subset of ML (AI) that uses ANNs with multiple layers (deep networks) to model complex patterns in data

→ [You only look once]
YOLO # Deep Convolutional Neural Network





Machine Learning

Vs

Deep Learning

*Po-tp

Feature	Machine Learning (ML)	Deep Learning (DL)
Definition	Algorithms that learn patterns from data without explicit programming.	Subset of ML that uses deep neural networks to model complex patterns.
Data Requirements	Works well with small to medium datasets.	Requires large datasets for better performance.
Feature Engineering	Requires manual feature extraction by domain experts.	Automatically extracts features from raw data.

Definition	programming.	Subset of ML that uses deep neural networks to model complex patterns.
Data Requirements	Works well with small to medium datasets.	Requires large datasets for better performance.
Feature Engineering	Requires manual feature extraction by domain experts.	Automatically extracts features from raw data.
Model Complexity	Uses simpler models like Decision Trees, SVM, and Random Forests.	Uses complex neural networks (CNNs, RNNs, Transformers).
Training Time	Faster training (minutes to hours).	Slower training (hours to weeks).
Computational Power	Works on CPUs; does not require high-end GPUs.	Requires GPUs/TPUs due to heavy computations.
Interpretability	More explainable and interpretable.	Harder to interpret (black-box nature).
Use Cases	Fraud detection, recommendation systems, structured data analysis.	Image recognition, speech processing, NLP, self-driving cars.
Generalization	Works well on structured/tabular data.	Works best with unstructured data (text, images, audio).
Cost	Lower cost due to lower hardware needs.	Higher cost due to expensive training and hardware.
Popular Libraries	Scikit-learn, XGBoost, LightGBM, H2O.ai.	TensorFlow, PyTorch, Keras, Theano.
Real-World Examples	Netflix recommendations, spam filtering, credit scoring.	Face recognition, self-driving cars, chatbots (ChatGPT).

APC to store a nice text book

→ unstructured data (text) ←

- ANN is a computational model inspired by the structure and function of the brain.

[there are billions of neurons]

- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

DL

Analysis

Here's the text from the image:

- ANN is a computational model inspired by the structure and function of the brain. (there are billions of neurons)
- ANN is the interconnected layers of neurons that process information and learn patterns from the data.

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LLM

(Large Language Models)

👤 Turing Test # Alan Turing → [Imitation Game]